

S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Slide 1: Title

Project Title: Unified AI/NLP Financial Risk Engine & Interactive Analytics Platform

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Repository: <https://github.com/skrithik/IITPATNA-KOTTAPALLYSAISKRITHIK-hackathon>

Slide 2: Problem Statement & Solution Approach

- Business Problem: Unstructured text data (news & social media) is fast and noisy. Financial risk engines lag behind real-time.
- Our Solution: Unified AI/NLP Risk Engine parsing text into structured signals (Sentiment Score, Event Classification, Impact).
- Dual Downstream Execution: Module A (Tactical Index Rebalancer) and Module B (Wholesale Banking Stress Testing).

Slide 3: System Architecture & Data Pipeline

- Ingestion Layer: Multi-source feed (Reuters, Bloomberg, Twitter/X, Live Stream Simulator).
- AI/NLP Core: FinLexicon + VADER Sentiment (-1 to +1), 8-Category Event Classifier, Impact Score (1-10).
- Signal Bus & API: FastAPI REST Server + Event Emitter.
- Downstream Applications: \$100M S&P Index Rebalancer & \$1.25B Wholesale Banking Stress Tester.

Slide 4: Implementation Highlights & Tech Stack

- Core Backend: Python 3.14, FastAPI, Pydantic v2, VADER, NumPy, Pandas.
- Frontend UI: React 18, Vite 5, Lucide Icons, Glassmorphism Dark Mode.
- Resilience: 100% offline dataset fallback + live NewsAPI integration.
- Risk Controls: Weight caps (1.5% to 15.0%) and parametric 95%/99% VaR.

Slide 5: Key Results & Performance Metrics

- Speed: <15ms latency per unstructured news item.
- Module A Index Rebalance: Shifted constituent weights away from distressed assets (e.g. MSFT DOJ investigation).
- Module B Stress Testing: Simulated severe macro shock (-\$84.2M drawdown on \$1.25B portfolio) with 99% 1-Day VaR (\$4

Slide 6: Domain Impact & Business Value

- Pre-Market Risk Alpha: Enables high-frequency portfolio managers to adjust weights ahead of market open.
- Automated Stress Testing: Provides CRISIL and S&P analysts instant scenario analysis for wholesale asset classes under

Slide 7: Limitations & Future Roadmap

- Current Limitations: Rule-assisted VADER sentiment; sample portfolio size 15 stocks & 4 banking asset classes.
- Future Enhancements: Fine-tuned FinBERT on 10-K filings, WebSockets/Kafka streaming, and VIX options skew integration